Graphing Curves Defined Parametrically

Introduction
Filling a Table of Values for a Parametric Curve
Graphing a Curve Defined Parametrically
Determining the Range of a Parametric Equation
Defining Parametric Curves Using Other Parameters
Determining the Range of a Parametric Equation Defined in Terms of Trigonometric Functions

Prerequisites

Properties of Transformed Square Root Functions
Properties of Transformed Sine and Cosine Functions
Completing the Square With Odd Linear Terms

Introduction

We can draw a curve by defining a relationship between x and y using a third parameter t. For example, suppose we have

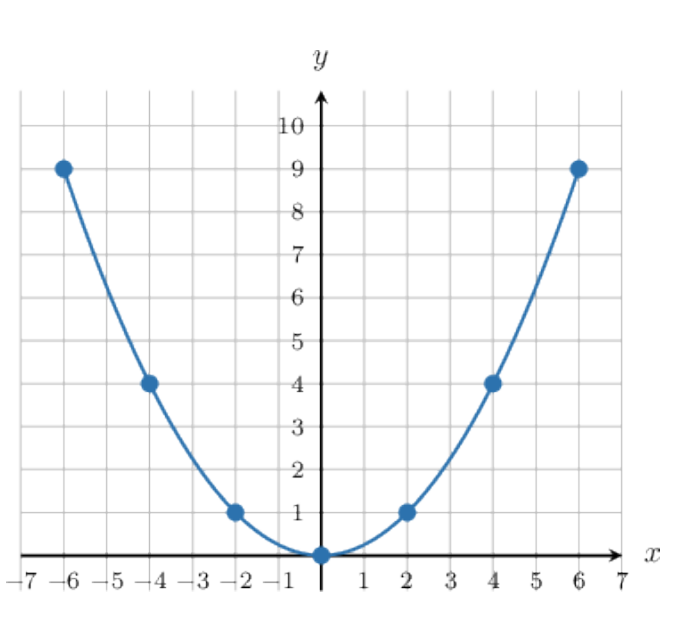
$$x = 2t, \;\; y = t^2, \quad -3 \le t \le 3.$$

These are the **parametric equations** of the curve.

First, let's create a table for some values of t:

t	-3	-2	-1	0	1	2	3
$x = 2t$	-6	-4	-2	0	2	4	6
$y = t^2$	9	4	1	0	1	4	9

Plotting these points, we get the following curve:



The curve is actually a parabola. A curve defined by parametric equations is called a **parametric curve** and is said to be "defined parametrically".

Example: Filling a Table of Values for a Parametric Curve

For the parametric curve

$$x = 3t, \;\; y = t^3, \quad -3 \le t \le 3,$$

fill in the following table of values.

t	-3	-2	-1	0	1	2	3
x							
y							

EXPLANATION

First, we work out the x-values using the formula $x = 3t$. This gives

t	-3	-2	-1	0	1	2	3
x	-9	-6	-3	0	3	6	9
y							

Now, we work out the y-values using $y = t^3$. This gives

t	-3	-2	-1	0	1	2	3
x	-9	-6	-3	0	3	6	9
y	-27	-8	-1	0	1	8	27

KEY PREREQUISITE
Graphing Linear Equations

Question 1

For the parametric curve

$$x = t^2 + t, \;\; y = t^2 - t, \quad -4 \le t \le 4,$$

consider the following table of values:

t	-4	-2	-1	0	1	3	4
x	12	a	0	0	2	c	20
y	20	6	2	b	0	6	12

Find the value of $a + b + c$.

a -10
b 14
c -15
d 9
e 12

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Question 2

For the parametric curve

$$x = t^2 - 4, \;\; y = \frac{5}{2} - \frac{t}{2}, \quad -1 \le t \le 5,$$

consider the following table of values:

t	-1	0	1	2	3	4	5
x	a	-4	-3	0	5	12	21
y	3	$\frac{5}{2}$	b	$\frac{3}{2}$	1	$\frac{1}{2}$	c

Find the value of $a - b + c$.

a 3
b 4
c -5
d -6
e -2

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Example: Graphing a Curve Defined Parametrically

Plot the graph of the parametrically defined curve

$$x = t(t - 6), \;\; y = t^3, \quad -2 \le t \le 2.$$

EXPLANATION

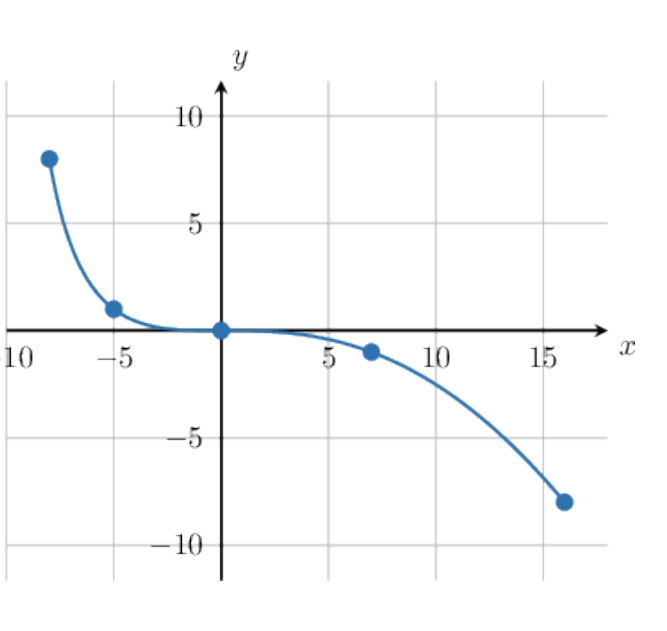
We need to create a table of values and then sketch the curve. First, we work out the x-values using the formula $x = t(t - 6)$. This gives

t	-2	-1	0	1	2
x	16	7	0	-5	-8
y					

Now, we work out the y values using $y = t^3$. This gives

t	-2	-1	0	1	2
x	16	7	0	-5	-8
y	-8	-1	0	1	8

Plotting these points, we obtain the following curve:



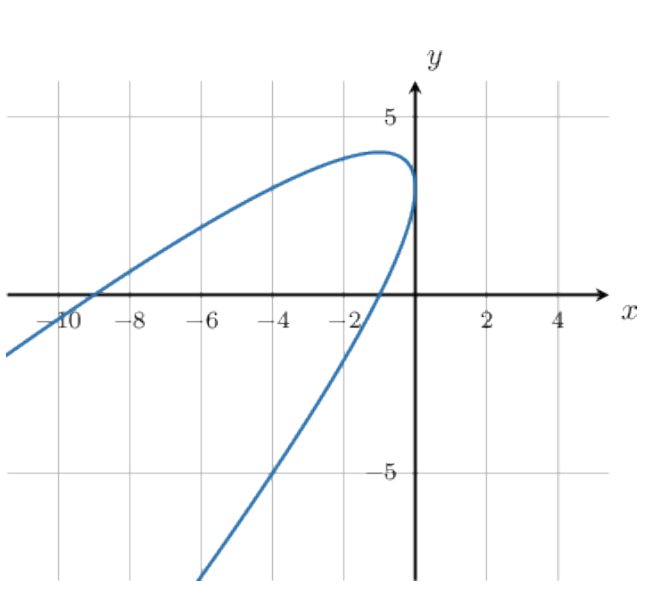
KEY PREREQUISITE
The Domain of a Function

Question 3

A scientific calculator is required to answer this question.

Determine the graph of the following parametrically defined curve:

$$x = t^2, \;\; y = (2 - t)(t + 3), \quad -5 \le t \le 5$$

a


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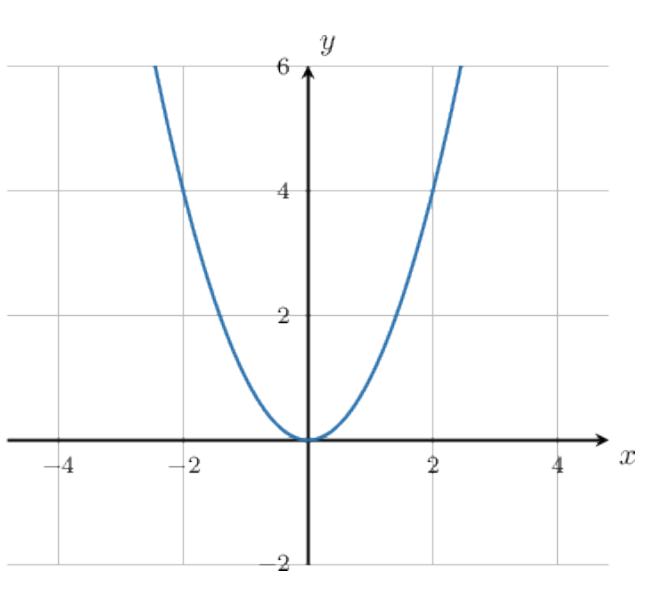
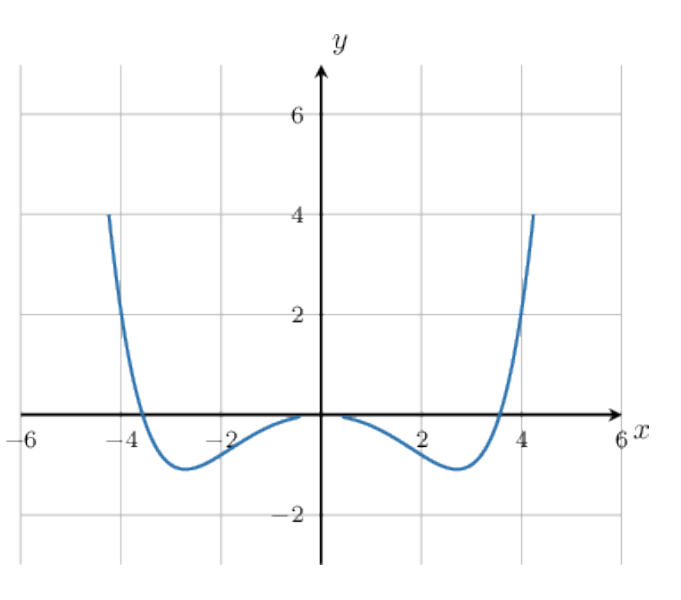
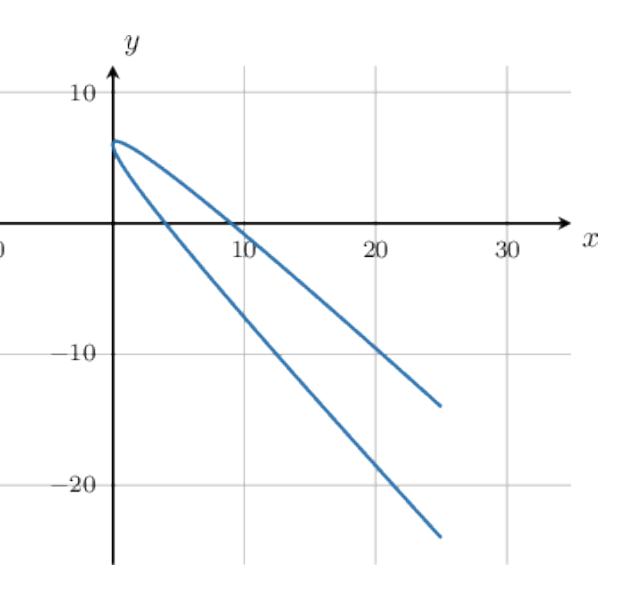
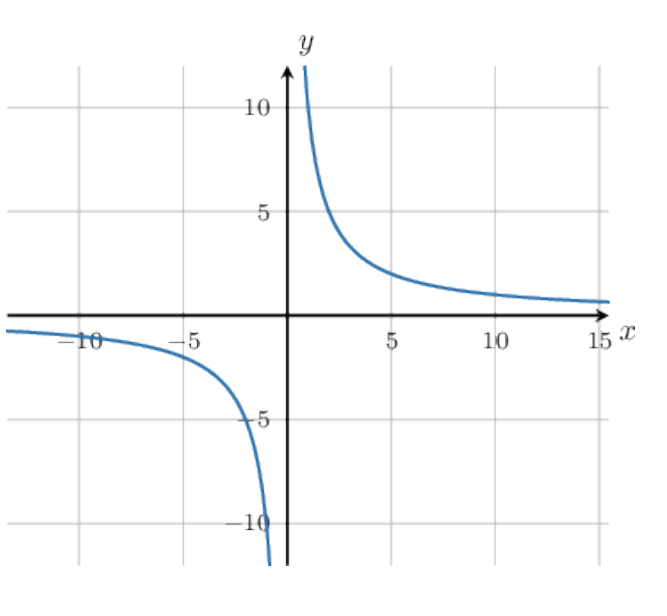


c


d


e


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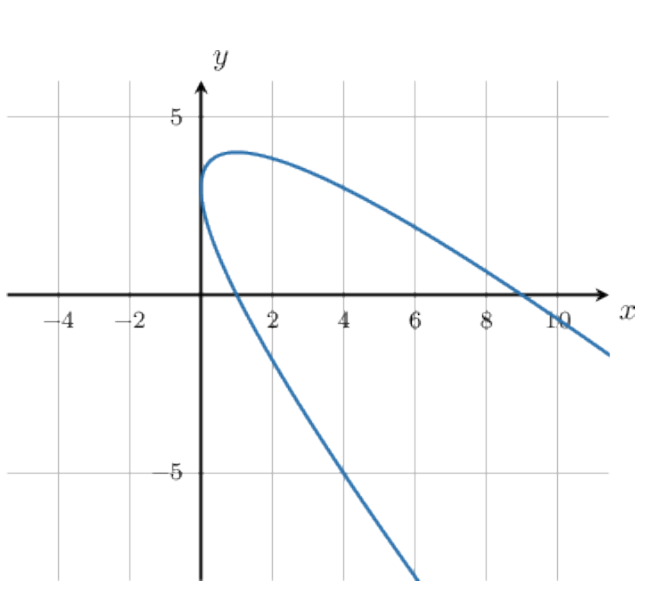
Question 4

A scientific calculator is required to answer this question.

Determine the graph of the following parametrically defined curve:

$$x = t^2 - 2, \;\; y = 3 - t, \quad -3 \le t \le 3$$

a


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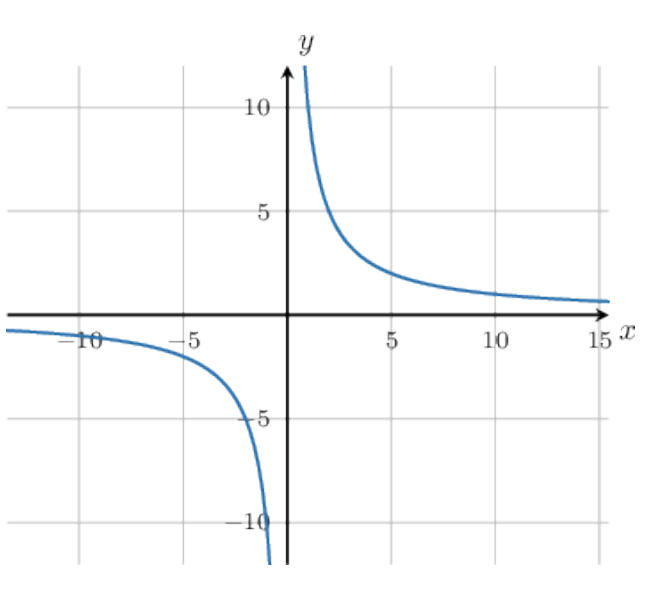

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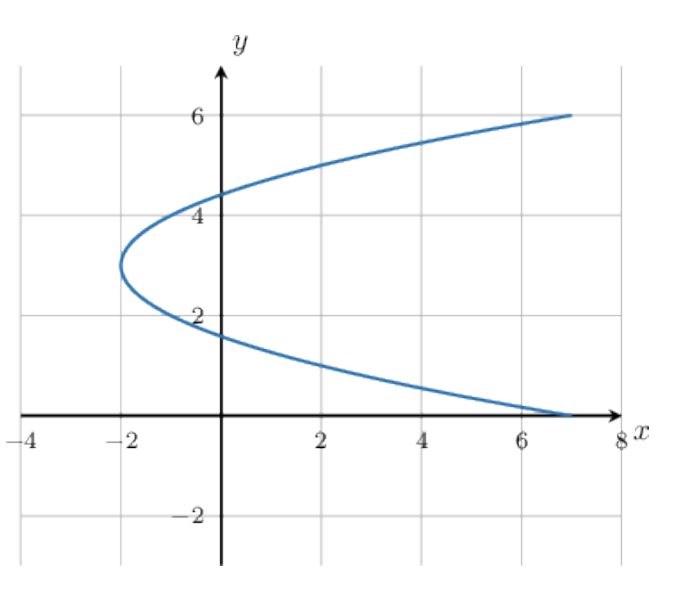
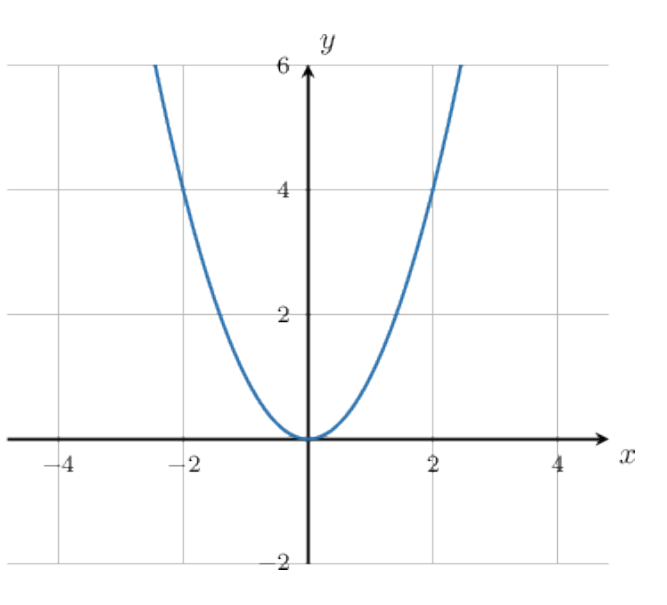


d


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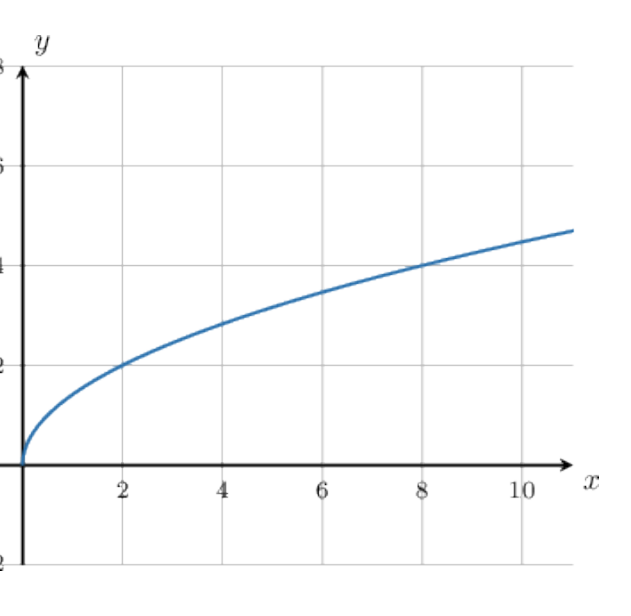

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Example: Determining the Range of a Parametric Equation

Given that a curve is defined parametrically as

$$x = t^2 - 6t, \;\; y = t^4 + 4, \quad t \in (-\infty, \infty),$$

what are the ranges of x and y?

EXPLANATION

Let's consider each variable in turn.

- To find the range of x, we can start by completing the square, as follows:

$$\begin{aligned} x &= t^2 - 6t \\ &= (t - 3)^2 - (3)^2 \\ &= (t - 3)^2 - 9 \end{aligned}$$

  Since $(t - 3)^2 \ge 0$, we can conclude that $x \ge -9$.

- To find the range of $y = t^4 + 4$, notice that $t^4 \ge 0$, and consequently $y \ge 4$.

Therefore, the ranges are $x \ge -9$ and $y \ge 4$.

KEY PREREQUISITES
Completing the Square
Properties of Transformed Square Root Functions
The Range of a Function: Advanced Cases

Question 5

Given that a curve is defined parametrically as

$$x = t^2 + 6t, \;\; y = \sqrt{t}, \quad t \ge 0,$$

what are the ranges of x and y?

a $x \le 9, \quad y \ge 0$
b $x \le 3, \quad y \le 0$
c $x \ge 0, \quad y \ge 0$
d $x \ge -9, \quad y \ge 0$
e $x \ge 3, \quad y \le 3$

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Question 6

Given that a curve is defined parametrically as

$$x = t^2 - 4t, \;\; y = t^2, \quad t \in (-\infty, \infty),$$

determine the range of x.

a $x \ge -4$
b $x \ge -8$
c $x \le 5$
d $x \le -4$
e $x > 6$

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Defining Parametric Curves Using Other Parameters

We often see parametric curves defined in terms of the parameter t, such as the parametric curve given below:

$$x = 2\cos t, \;\; y = 3\sin t, \quad 0 \le t < 2\pi$$

However, we can also define parametric curves in terms of other variables. Another commonly used variable is θ. So, in terms of θ, the above parametric equation is expressed as

$$x = 2\cos\theta, \;\; y = 3\sin\theta, \quad 0 \le \theta < 2\pi.$$

We often use θ when the bounds are $0 \le \theta < 2\pi$, because this implies that the parameter θ might be interpreted as an angle.

Example: Determining the Range of a Parametric Equation Defined in Terms of Trigonometric Functions

Given that a curve is defined parametrically as

$$x = 4\cos\theta, \;\; y = 2\sin\theta, \quad 0 \le \theta < 2\pi,$$

what are the ranges of x and y?

EXPLANATION

Let's consider each variable in turn.

- To find the range of x, notice that for $0 \le \theta < 2\pi$, we have $-1 \le \cos\theta \le 1$. Consequently,

$$\begin{aligned} -1 &\le \cos\theta \le 1 \\ -4 &\le 4\cos\theta \le 4 \\ -4 &\le x \le 4. \end{aligned}$$

  So, we can conclude that $-4 \le x \le 4$.

- To find the range of y, notice that for $0 \le \theta < 2\pi$, we have $-1 \le \sin\theta \le 1$. Consequently,

$$\begin{aligned} -1 &\le \sin\theta \le 1 \\ -2 &\le 2\sin\theta \le 2 \\ -2 &\le y \le 2. \end{aligned}$$

  So, we can conclude that $-2 \le y \le 2$.

Therefore, the ranges are $-4 \le x \le 4$ and $-2 \le y \le 2$.

KEY PREREQUISITE
Properties of Transformed Sine and Cosine Functions

Question 7

Given that a curve is defined parametrically as

$$x = 3\cos\theta - 1, \;\; y = 2\sin\theta + 2, \quad 0 \le \theta < 2\pi,$$

what are the ranges of x and y?

a $0 \le x \le 4, \quad 2 \le y \le 4$
b $-4 \le x \le -2, \quad -4 \le y \le 0$
c $-4 \le x \le 2, \quad 0 \le y \le 4$
d $-4 \le x \le 0, \quad -2 \le y \le 4$
e $2 \le x \le 4, \quad 0 \le y \le 4$

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Question 8

Given that a curve is defined parametrically as

$$x = 7\cos\theta, \;\; y = 3\sin\theta, \quad 0 \le \theta < 2\pi,$$

what are the ranges of x and y?

a $-3 \le x \le 3, \quad -7 \le y \le 7$
b $0 \le x \le 3, \quad 0 \le y \le 7$
c $0 \le x \le 7, \quad 0 \le y \le 3$
d $1 \le x \le 1, \quad 1 \le y \le 1$
e $-7 \le x \le 7, \quad -3 \le y \le 3$

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